Atharv Pratap Singh Chauhan

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Education

Indian Institute of Technology (IIT) Bombay Mumbai, India
B.Tech in Mechanical Engineering | CGPA: 8.39 *Jul 2025 – Present*

The University of Tokyo Online
Global Consumer Intelligence (GCI) Data Science Program *Mar 2026 – Present*

Experience

Indian Institute of Technology (IIT) Bombay Mumbai, India
Undergraduate Research Assistant (SURP) | Prof. Debabrata Maiti *May 2026 – Present*

- Selected for the Summer Undergraduate Research Program (SURP) to research **Brain-Inspired Neural Networks for Multi-Modal Learning and Representation**.
- Focusing on designing data-efficient, robust deep learning architectures by mimicking biological principles such as synaptic plasticity, modularity, and hierarchical processing.
- Developing a rigorous mathematical foundation in machine learning by completing **Stanford's CS229** curriculum.
- Utilizing algorithmic optimization and high-performance Python programming to translate theoretical neuroscience frameworks into functional computational models.

Awards & Achievements

Algo General Championship (IIT Bombay) | *Rank 5 Overall* Apr 2026

- **Secured 5th position institute-wide** across all undergraduate years in the 3-hour algorithmic contest hosted by WnCC, successfully solving **5 out of 6 problems**.

Jane Street Monthly Puzzle | *Global Top 50 Solver* Mar 2026

- **Ranked 42nd globally** for providing a rigorous, correct submission to the March 2026 puzzle ("Planetary Parade").

Codeforces | *Candidate Master* Mar 2026

- Currently **ranked top 10** among active competitive programmers at the institute (**Peak Rating: 1957**).

IMC Pit Trading Challenge | *3rd Place* Jan 2026

- **Secured 3rd position out of 200+ teams** in the challenge hosted by Quant Community, IIT Bombay and IMC Trading.

HackerRank | *5-Star Coder* Oct 2025

- Achieved **5-star proficiency** in C++ and Python, with a global rank in the **top 1.4% of over 4 million users**.

Projects

Real Time HFT Execution Engine | *C++, Python, Matplotlib* Dec 2025 – Jan 2026

- Engineered a low-latency trading simulation engine bridging high-performance system design with real-time visualization.
- Built the core C++ exchange and execution layer handling market data (Random Walk) and trade logic (Moving Average Crossover) with **sub-millisecond latency**.
- Designed a **Producer-Consumer IPC model** using standard OS pipes to stream data efficiently between the C++ backend and Python frontend.
- Developed a live financial dashboard featuring a thread-blocking pause/resume mechanism and dynamic P&L tracking with **automated Max Drawdown calculations**.

Stochastic Risk Engine | *C++, Python, Pandas* Dec 2025

- Designed a high-performance engine to model Geometric Brownian Motion, bridging low-latency C++ computation with Python data pipelines.
- Engineered a multi-threaded C++ core capable of handling **100+ concurrent market simulations** using custom random number generation and optimized File I/O.
- Implemented Value at Risk (VaR) algorithms using **Drift + Shock mathematical modeling** to quantify potential portfolio losses at a 95% confidence interval.
- Architected a visualization layer rendering the "Cone of Uncertainty" for asset price predictions.

Technical Skills

Languages: C++, Python
Frameworks & Libraries: Django, Matplotlib, Pandas
Relevant Coursework: Calculus, Linear Algebra & Differential Equations, Computer Programming Utilization
Tools & Platforms: Git, GitHub
Core Competencies: Data Structures & Algorithms, Quantitative Analysis, Mathematical Modeling